Zipeng Wu

Time-Series Representation and Downstream Applications | ICML 2026 First Author

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PhD Awards and Funding

College PhD scholarship	University of Birmingham College PhD scholarship, Jan 2024–Jul 2027: **full scholarship support** for Applied Mathematics PhD study, covering **tuition fees and stipend/living costs.**
UKRI/AIRR 2026	*From Minutes to Decades: Learning World Models of Human Ageing from the Wrist* (AIRR-GW 0261-3202-4325-1), Oct 28 2026-Jan 26 2027: **10,000 GPU-hours** on Isambard-AI; nominal compute-resource value **£45,000**.
UKRI/AIRR 2026	*Language-Action Time-Series Tokenization for Efficient VLA Policies* (AIRR-GW 0261-9925-2489-1), Jul 1–Sep 29 2026: **10,000 GPU-hours** compute allocation on Isambard-AI, with nominal compute-resource value **£45,000**; compute resources only, not direct cash funding.
UKRI/AIRR 2026	*Time Series Language and Foundation Model* (AIRR-GW 0261-5623-3483-1), Jun 11–Sep 9 2026: **10,000 GPU-hours** compute allocation on Isambard-AI, with nominal compute-resource value **£45,000**; compute resources only, not direct cash funding.
3 awards	**Three approved UKRI/AIRR compute awards total 30,000 GPU-hours on Isambard-AI; nominal resource value £135,000; compute resources only, not direct cash funding.**
Contribution	Authored all three proposals and conducted most of the experimental work.
UK Biobank	Transfer funding grant: **£1,000**.

Publications and Presentations

Preprint 2026	**WISP: Composable Representations for Wrist-Worn Activity Recognition.** Zipeng Wu, Shiqiao Zhou, Fabian Spill, and James Andrews. Preprint, 2026. First author. [Pure record]
ICML 2026	**Time-Series Decomposition as a Standalone Task: A Mechanism-Driven Diagnostic Benchmark. Zipeng Wu**, Jiani Wei, Shiqiao Zhou, Jiajun Chen, Fabian Spill, and J. W. Andrews*. *Forty-third International Conference on Machine Learning (**ICML 2026**)*, **CORE/ICORE A***, 2026. **First author**.
KDD 2026	**Stationarity-Aware Retrieval-Augmented Time Series Forecasting.** Shiqiao Zhou, Holger Schöner, **Zipeng Wu**, Edouard Fouché, IAG Wilson, and Shuo Wang*. *32nd **SIGKDD** Conference on Knowledge Discovery and Data Mining (**KDD 2026**)*, **CORE/ICORE A***, 2026. Co-author.
BIBM 2024	**iTARGET: Interpretable Tailored Age Regression for Grouped Epigenetic Traits. Zipeng Wu**, Daniel Herring, Fabian Spill, James Andrews. *IEEE International Conference on Bioinformatics and Biomedicine (**BIBM**)*, **CORE/ICORE B**, pp. 647–652. **First author**.
Heliyon 2023	**A novel online multi-task learning for COVID-19 multi-output spatio-temporal prediction. Zipeng Wu**, Chu Kiong Loo, Unaizah Obaidellah, Kitsuchart Pasupa. *Heliyon*, **JCR Q1**, 9(8), e18771, 2023. **First author**.
ICONIP 2023	**Correlated Online k-Nearest Neighbors Regressor Chain for Online Multi-output Regression. Zipeng Wu**, Chu Kiong Loo, Kitsuchart Pasupa. *ICONIP 2023*, **CORE/ICORE B**, LNCS 14449, pp. 28–39. **First author; oral presentation**.
ICONIP 2022	**An Interpretable Multi-target Regression Method for Hierarchical Load Forecasting. Zipeng Wu**, Chu Kiong Loo, Kitsuchart Pasupa, Licheng Xu. *ICONIP 2022*, **CORE/ICORE B**, CCIS 1794, pp. 3–12. **First author; oral presentation**.
IJCNN 2020	**A Novel Dynamically Adjusted Regressor Chain for Taxi Demand Prediction. Zipeng Wu**, Guan Lian. *International Joint Conference on Neural Networks (IJCNN)*, **CORE/ICORE B**, 2020. **First author; oral presentation**.

Selected Health-related Research Projects (Unpublished)

Epigenetic Ageing Models (Ongoing)
Developed interpretable ageing models from DNA methylation data across multinational cohorts, evaluating cross-study generalisation through independent validation.

Multimodal Ageing and Health Modelling (Ongoing; manuscript in preparation)
Developed multimodal models using UK Biobank wearable, cardiac and brain-derived data to investigate biological ageing and longitudinal health outcomes.

Wearable-based Longitudinal Health Modelling (Ongoing; manuscript in preparation)
Extended UK Biobank-derived wearable modelling approaches to the English Longitudinal Study of Ageing (ELSA) and the US National Health and Nutrition Examination Survey (NHANES), with benchmarking against published evidence from the US Lifestyle Interventions and Independence for Elders (LIFE) trial.

Education

Jan 2024-present
Expected award:
Jul 2027-Jan 2028

PhD Researcher in Applied Mathematics — University of Birmingham, UK
Supervisors: Dr James Andrews; Prof Fabian Spill (UKRI Future Leaders Fellow).
Focus: time-series representation and downstream applications, spanning decomposition, similarity/retrieval and prediction/regression.

2020–2023

Master of Computer Science (Research) — Universiti Malaya, Malaysia
Research in online machine learning, multi-output regression, spatio-temporal prediction, and time-series forecasting.

2015–2019

Bachelor of Engineering — Guilin University of Electronic Technology, China
Information Technology; computing and engineering-oriented problem solving.

Open Source

TOMC Memory CPU-based context preparation and persistent task notes for long-running agents. [GitHub]

De-Time Python/CLI time-series decomposition toolkit with components and metadata. [GitHub] [Web] [Demo]

EchoTime Explainable structural time-series similarity; JSON output and HTML reports. [GitHub] [Web]

TSDecompose-Benchmark Mechanism-driven decomposition-quality benchmark (ICML 2026). [GitHub] [Data]

WristHARBench Harmonised wrist-HAR benchmark with participant-held-out evaluation. [Hugging Face]

Quantitative Research Internships

Feb–Jul 2021

Quantitative Strategist Intern — Endemaj Token Fund, Kuala Lumpur
Developed digital-asset quantitative strategies, exchange-trading prototypes, and Python backtesting workflows for strategy evaluation.

Jun–Sep 2018

Quantitative Strategist Intern — Beijing Singularity Disintegration Co., Beijing
Worked with cryptocurrency OHLCV/minute data; built backtests using Sharpe ratio, maximum drawdown, moving averages, RSI, and Bollinger-band-style signals.

Technical Skills

HPC & distributed training: Hands-on use of **Isambard-AI** (**5,448-GPU system**); **Megatron** foundation-model training.
Research engineering: Python, PyTorch, NumPy, SciPy, pandas, scikit-learn, Git/Linux; reproducible pipelines and benchmarks.

Research Competition

Biomarkers of Aging Challenge 2024 - Chronological Age Prediction.
Leaderboard rank 16; final evaluation MAE 3.079 years. [Leaderboard]